30 June 1955

SUPERSEDING
NAVORD OS - 1366 B
5 FEBRUARY 1945

MILITARY SPECIFICATION

BANDS, PROJECTILE ROTATING

1. SCOPE

1.1 Scope.- This specification augments pertinent drawings and specifications shown on the list of drawings and specifications particular to the item of the contract so that a study of the contract, drawings and specifications will clearly set forth the manufacturing and inspection requirements for projectile rotating bands which, before assembly are blanks in the form of rings of drawn, forged or rolled material.

1.2 Classification.- Projectile rotating bands shall be of the following materials, as specified in Table I.

Copper
Guilding Metal
Cupro-Nickel

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-16594 Projectiles, Bombs, Rockets, and Guided Missile Warheads, Inert Parts, Manufacture of

NAVY DEPARTMENT

General Specifications for the Inspection of Material, Appendix II, Part A

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

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PUBLICATIONS:

NAVY DEPARTMENT

OP400 General Instructions for the Design, Manufacture, and Inspection of Naval Ordnance Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The material shall be as specified on the applicable projectile drawing.

3.1.1 Chemical composition and physical requirements.- Chemical composition and physical requirements shall be as specified in Tables I and II.

TABLE I

Chemical Composition

Material	(1) Copper (percent)	Iron (maximum percent)	Lead (maximum percent)	Other elements (percent)	Total impurities (maximum percent)
Copper	99.88(2)	-	-	-	0.12
Gilding Metal	89-93	0.05	0.05	Zinc(3)	0.125
Cupro-Nickel	(3)	0.10	-	Nickel 2.0-3.0 Manganese 0.40	0.30(4)

(1) Any silver present shall be counted as copper.
(2) Minimum.
(3) Remainder.
(4) Elements not otherwise specified.

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TABLE II

Physical Properties (from tensile tests)

Material	Caliber of Projectile (inches)	Tensile Strength (minimum, lbs. per sq. inch)	Elongation (minimum percent) In 1 inch	Elongation (minimum percent) In 2 inches	For Test specimen, see	Condition
Copper	Less than 1.0	30,000	30	--	4.3.2	(1)
	1.0 to 1.875	30,000	30	--	4.3.2.2	(1)
	1.88 to 4.50	30,000	--	15	4.3.2.2	(2)
	4.51 to 5.50	30,000	--	25	4.3.2.2	(2)
	5.51 to 7.50	30,000	--	30	4.3.2.2	(2)
	Over 7.50	30,000	--	30	4.3.2.2	(2)
Gilding Metal	Less than 1.0	32,000	30	--	4.3.2	(1)
	1.0 to 1.875	32,000	30	--	4.3.2.2	(1)
	1.88 to 4.50	35,000	--	15	4.3.2.2	(2)
	4.51 to 5.50	35,000	--	25	4.3.2.2	(2)
	Over 5.50	35,000	--	30	4.3.2.2	(2)
Cupro-Nickel	Over 7.50	30,000	--	30	4.3.2.2	(2)

(1) - Test specimens taken from annealed blanks before assembly.
(2) - Test specimens taken from bands removed from projectiles after assembly, but before machining.

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3.2 Assembly

3.2.1 Annealed blanks -- Projectile rotating band blanks shall be in the annealed condition when assembled.

3.2.2 Cold blanks - Blanks for projectiles smaller than 6-inch shall be assembled cold.

3.2.3 Seating - Band seat shall be free of all foreign material prior to assembly of the band. Blanks shall be pressed to a tight fit in the roughened band seat in such a manner as not to deform the projectile body. The assembled band shall completely fill the band seat.

3.2.4 Heated blanks - Blanks for 6-inch and larger projectiles may be heated and assembled hot, provided the blank is not heated to a temperature above 1650°F and provided the furnace is fitted with pyrometric control to insure uniformity of assembly. Blanks assembled hot shall be allowed to cool to at least 750°F under the last application of banding pressure. Care shall be taken to prevent oxidation during heating. Blanks which are heated for hot assembly to a temperature higher than the annealing temperature need not be annealed before being heated for assembly.

3.3 Bending - Bands shall withstand the bending test specified in 4.3.3 without cracking on the outside of the bent portion.

3.4 Workmanship - All blanks shall be free from slivers, pipes, laps, cracks, laminations, grooves, seams, oxidation flaws, deep scratches, and other defects which may render them unsuitable for the purpose intended. All blanks shall be round, of uniform dimensions, and free from scale, chips, oil, grease, and dirt.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Sampling.- Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government Inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Inspection Lot.- As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot," i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.2 Production Lot.- Unless otherwise specified each production lot of assembled bands shall be the same quantity as required by the applicable projectile specification.

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4.1.3 Sampling for destructive tests.- At least one band shall be carefully removed from each production lot, before machining, to permit examination of seating and physical tests. It is to be expected that several times this minimum number will need to be selected at the start of production; as production progresses the number may be reduced. The choice of the number of samples to be selected from any lot shall lie with the Inspector.

4.2 Inspection.- The general inspection procedures shall be in accordance with General Instructions For the Design, Manufacture, and Inspection of Naval Ordnance Equipment, OP 400; Government Inspection shall be in accordance with the provisions of General Procedure for Government Inspection and Contractor's Inspection shall be in accordance with Inspection by Contractor, as contained in OP 400.

4.3 Test Procedures

4.3.1 Seating.- The seating of assembled bands shall be tested before machining by tapping the band lightly with a steel hammer. A hollow sound accompanied by an unusual rebound of the hammer indicates an improperly seated band. A more critical test may be used for the examination of doubtful cases such as laying a small coin (a dime of U. S. currency) on the band and tapping the band near the coin with a 4-ounce hammer. An unseated condition under the coin will be shown by a marked rebound of the coin.

4.3.1.1 At least one band, and a greater number as the Inspector considers necessary, shall be carefully removed from each lot of projectiles to permit examination of the seating. If there is an indication that the assembly has not been performed satisfactorily, all projectiles in the lot shall be rebanded. Projectiles with improperly seated bands and those from which bands have been removed may be rebanded and resubmitted for test.

4.3.2 Tension test.- For blanks with an inside diameter less than 1-inch, a sample of the tubes from which the blanks are cut shall have the full section tested in accordance with Appendix II, Metals, referred to in 2.1. If blanks are cut from hard tubing and then annealed, the test section shall be annealed with the blanks which it represents.

4.3.2.1 For blanks with an inside diameter of 1-inch or more, tension test specimens shall be cut circumferentially from the annealed blanks or from bands which have been removed after assembly as prescribed in Table II. The specimens shall be straightened cold by pressure, without hammering, and otherwise prepared in a manner as to minimize cold working.

4.3.2.2 Test specimens for various sizes shall be as follows:

If the blank has an inside diameter less than 7.50 inches but not less than 1.0 inch, in accordance with figure 1 of this specification.

If the blank has an inside diameter of 7.50 inches or larger, in accordance with type 5 specimen of Appendix II, Metals, Part A, referred to in 2.1.

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The thickness of test specimens shall be the full thickness of the material, except that a small amount may be removed by machining when necessary to obtain a suitable test specimen.

4.3.3 Bending test.- For all sizes, the bending test shall be made on specimens taken from bands removed from projectiles after assembly but before machining. Jaw marks and similar surface imperfections may be removed by machining and sharp corners may be "broken" with a smooth file.

4.3.3.1 Small bands.- In case of small bands, bending test specimens shall consist of whole bands cut open on one side.

4.3.3.2 Large bands.- In case of large bands, bending test specimens shall consist of a section, of the full width and thickness, with length approximately one-third the circumference.

4.3.3.3 Procedure.- The specimen shall be bent cold by pressure, without hammering, in the direction of curvature of the blank, until it tightly encloses 180° of a rod equal in diameter to the thickness of the specimen.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.- Projectile rotating band blanks shall be packaged and packed in accordance with the contract requirements.

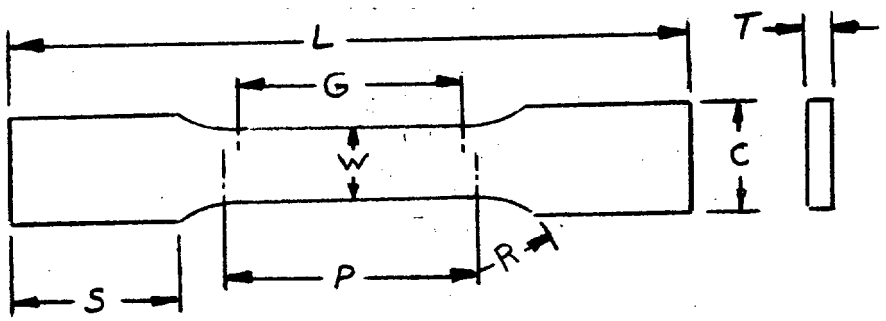
6. NOTES

6.1 It is to be noted that proper seating of a band so that it is in contact with the band seat at all points is an important factor in ballistic dispersion and retention of band.

6.2 All correspondence, instructions, or practices that are in conflict with and are dated or in use prior to the latest revision of this specification are hereby cancelled.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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	Inside diameter (inches)	
	1 to 1.875	Over 1.875 to 7.5, exclusive
	Inches	Inches
G (gage length) ---------	1.0	2.0
L approximate ----------	(1)	9.0 (2)
P minimum ---------------	1.125	2.25
R (radius), minimum -----	.50	.50
S approximate ----------	(3)	2.0 (3)
T (4) ------------------	(5)	(5)
W ----------------------	0.500 ± 0.010	0.500 ± 0.010
C (6) approximate -------	.625	.625

(1) Maximum obtainable.
(2) These dimensions shall be maximum obtainable when the flattened blank is less than 9 inches long.
(3) Remainder of blank.
(4) See paragraph 4.3.2.2
(5) Band thickness.
(6) When blank is 1/2 inch to 5/8 inch, inclusive, in width, the full width of the blank shall be used.

FIGURE I

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